Laboratory work 4 - Approximations of Bode magnitude diagrams by asymptotes

Bode diagram

A **Bode diagram** is a graphical representation of a system's frequency response, how it behaves across a wide range of frequencies. It consists of two graphs plotted against frequency (ω) on a logaritmic scale:

1. **Magnitude plot**: shows how much the system **amplifies or attenuates** signals at different frequencies, expressed in **decibels (dB)**

2. **Phase plot**: shows the **phase shift (in degrees or radians)** as frequency changes

From previous laboratory works we can remember what is the cut-off frequency, what represents the corner frequencies and how we can convert to logaritmic scale.

In this laboratory, it is presented how we can approximate bode diagrams using straight-line asymptotes. To be able to start, we should know how the slope changes according to the type of singularity in the transfer function.

Real negative pole : $\frac{1}{Ts+1}$ → $\omega_{co} = \frac{1}{T}$

LF : 0 dB/dec slope
HF : -20 dB/dec slope

Phase : $(0°, -90°)$
$-45°$ at ω_{co}

Real negative zero : $Ts+1$ → $\omega_{co} = \frac{1}{T}$

LF : 0 dB/dec slope
HF : +20 dB/dec slope

Phase : $(0°, +90°)$
$+45°$ at ω_{co}

Pair of complex pole : $\frac{1}{\frac{s}{\omega_m^2}+\frac{2\zeta s}{\omega_m}+1}$ → $\omega_{co} = \omega_m$

LF : 0 dB/dec slope
HF : -40 dB/dec slope

Phase : $(0°, -180°)$
-90 at ω_{co}

Pair of complex zero : $\frac{s}{\omega_m^2}+\frac{2\zeta s}{\omega_m}+1$ → $\omega_{co} = \omega_m$

LF : 0 dB/dec slope
HF : +40 dB/dec slope

Phase : $(0°, +180°)$
$+90$ at ω_{co}

Pole in origin : $\frac{1}{s}$ → $\omega_{co} = 1$

-20 dB/dec slope
0dB at $\omega=1$

Phase : $-90°$

Zero in origin : s → $\omega_{co} = 1$

+ 20 dB/dec slope
0 dB at $\omega=1$

Phase : $+90°$

ω	$\approx \lg \omega$
1	0
2	0.3
3	0.5
4	0.6
5	0.7
6	0.8
7	0.85
8	0.9
9	0.95
10	1

To compute such an approximation, we have to follow some steps.

Problem

Example: i) H(s) = 20$\frac{s+2}{(s+10)(s+5)}$

Steps:

1. Transfer function - standard form: H(s) = k$\frac{(\mathrm{T1s}+1)}{(\mathrm{T2s}+1)}\ldots$

H(s) = $20\frac{2\left(\frac{1}{2}s+1\right)}{10\left(\frac{1}{10}s+1\right)5\left(\frac{1}{5}s+1\right)}$ = $\frac{4}{5}\frac{\left(\frac{1}{2}s+1\right)}{\left(\frac{1}{10}s+1\right)\left(\frac{1}{5}s+1\right)}$

```
H = tf(20*[1 2], conv([1 10],[1 5]));
zpk(H)
```

```
ans =

    20 (s+2)
  ------------
  (s+10) (s+5)

Continuous-time zero/pole/gain model.
Model Properties
```

2. Place the corner frequencies in the ascending order and mention the contribution to the slope: ω = $\frac{1}{\mathrm{T}}$

ω_2 = 2 (+20dB), ω_3 = 5 (-20dB), ω_4 = 10 (-20dB)

3. Range of the frequencies: ± 1 decade before and after smallest/biggest

Range = ($10^{-1}, 10^{2}$)

```
w = [1e-1, 2, 5, 10, 1e2]; % vector that contains the frequencies in the ascending
order + range margins
```

4. Compute the approximation: for this step we will use semilogx function

- from 10^{-1} to ω_2, the magnitude is $|k|^{\mathrm{db}}$

```
k = 4/5; kdB = 20*log10(k); % converting the proportional gain in decibels
```

- from ω_2 to ω_3, a slope of +20dB occurs

```
kdiff23 = 20*log10(w(3)/w(2));

% kdiff is represented as the distance between two frequencies on a logarithmic
scale converted in decibels/decade (slope)
```

```
% => change in magnitude over a specific frequency interval (kdiff = slope*lg(wi+1/
wi))
```

- from ω_3 to ω_4, the slope induced is -20dB => the slope becomes 0dB

```
% kdiff34 = kdiff23; because the slope is 0dB => straight line
```

- from ω_4 to 10^2, the slope induced is -20dB => the slope becomes -20dB

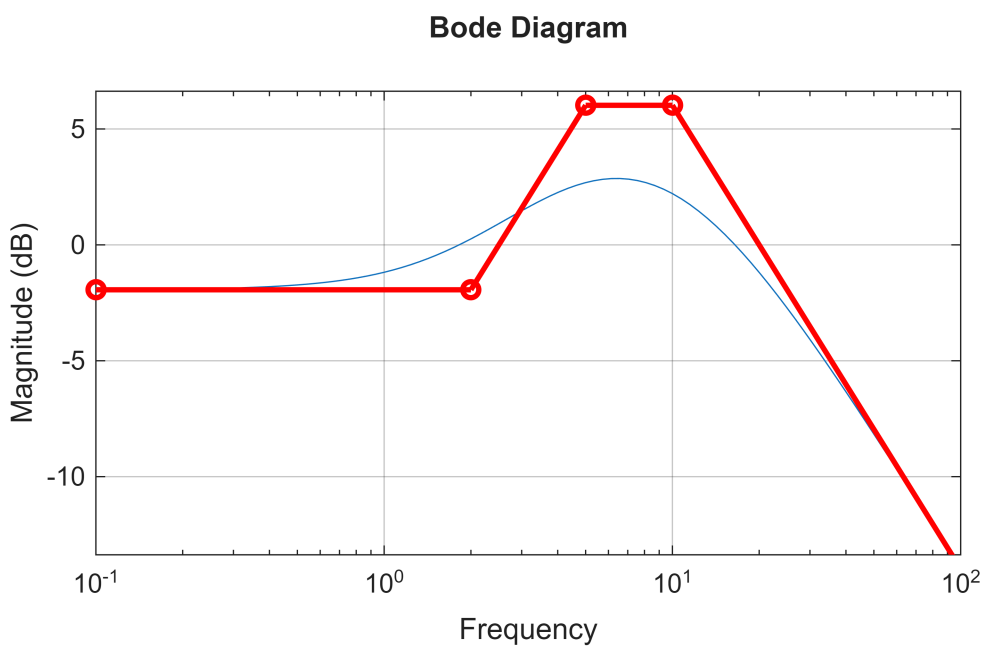
```
kdiff45 = -20*log10(w(5)/w(4));
```

5. Verify the accuracy using bodemag function

```
wl = logspace(-1, 2, 1e2); % we can specify on which decades the bode should be
represented
bodemag(H, wl);
hold; grid; shg;
```

```
Current plot held
```

```
semilogx([w(1), w(2)], [kdB, kdB], '-o', 'LineWidth', 2, 'Color', 'r');
semilogx([w(2), w(3)], [kdB, kdB + kdiff23], '-o', 'LineWidth', 2, 'Color', 'r');
semilogx([w(3), w(4)], [kdB + kdiff23, kdB + kdiff23], '-o', 'LineWidth', 2,
'Color', 'r');
semilogx([w(4), w(5)], [kdB + kdiff23, kdB + kdiff23 + kdiff45], '-o', 'LineWidth',
2, 'Color', 'r');
hold;
```




```
Current plot released
```

In order to do this computation easier, I created the "bode_approx" function at the laboratory with the input parameters: H, k, w, s and wl; where ***H*** is the transfer funtion of the system***, k*** is the proportional gain of the

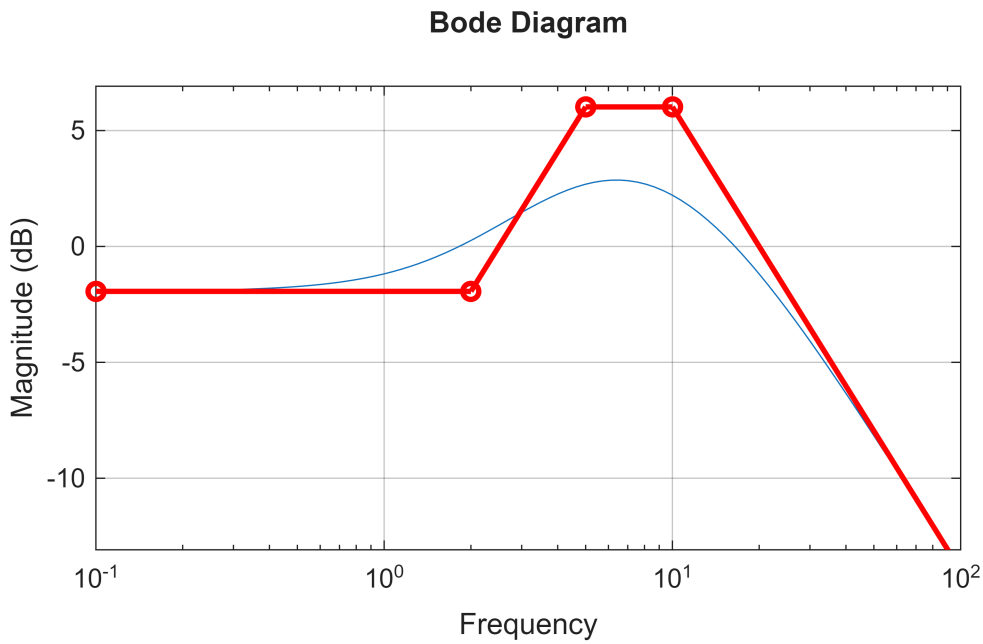
transfer function, ***w*** is a vector containing the frequencies in the ascending order, ***s*** is a vector with the slopes after each corner frequency and ***wl*** which represents the range on which the bode is plotted. This function consists of a for loop and help us generalize the approximation process.

```
function [] = bode_approx(H, k, w, s, wl)

    kdB = 20*log10(k) + s(1)*log10(w(1)/1);
    % a pole/zero in the origin gives +-20 slope line => we shift it to the
starting frequency
    bodemag(H, wl); grid; shg; hold;

    for i = 1:length(w)-1
        % the loop repeats length(w)-1 times because in each current step we call
the next value in the frequency vector
        kdbdiff = s(i)*log10(w(i+1)/w(i)); % computing the change in magnitude
        semilogx([w(i), w(i+1)], [kdB, kdB + kdbdiff], '-o', 'LineWidth', 2,
'Color', 'r');
        kdB = kdB + kdbdiff; % update the magnitude with the ending point of the
previous asymptote
    end

    hold;

end
```

And now, let's apply this function on the same example.

```
H = tf(20*[1 2], conv([1 10],[1 5]));
w = [1e-1, 2, 5, 10, 1e2]; % vector that contains the frequencies in the ascending
order + range margins
s = [0, 20, 0, -20]; % vector with the slopes after each corner frequency
k = 4/5;
wl = logspace(-1, 2, 1e2); % margins for the bode plot
bode_approx(H, k, w, s, wl);
```

```
Current plot held
Current plot released
```

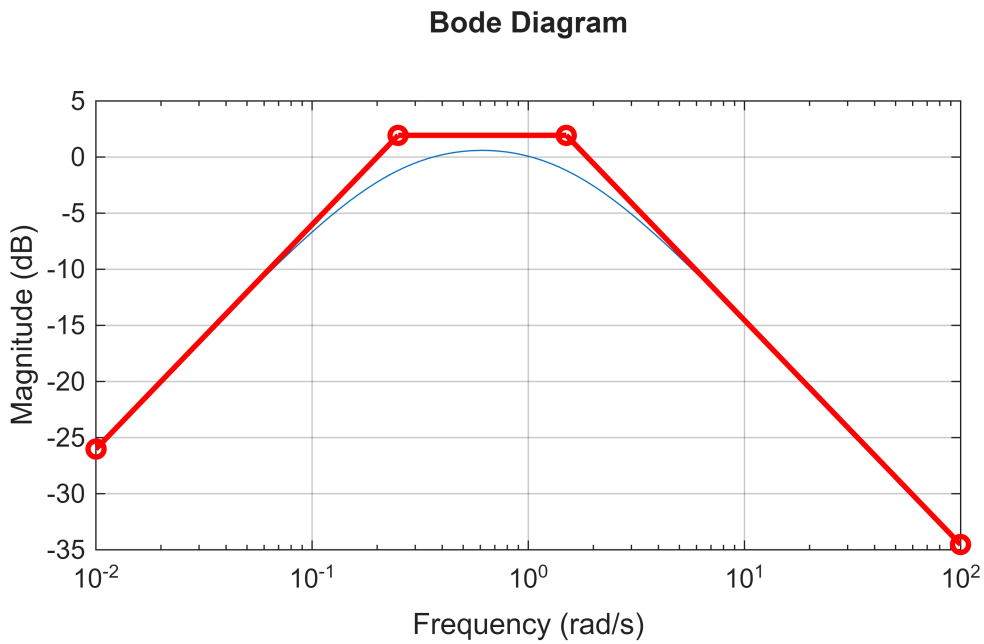


And now let's try with a transfer function that contains a zero in the origin.

H(s) = $\frac{15s}{(2s+3)(4s+1)} = \frac{15}{3}\frac{s}{\left(\frac{2}{3}s+1\right)(4s+1)} = 5\frac{s}{\left(\frac{2}{3}s+1\right)(4s+1)}$

```
H = tf(15*[1 0], conv([2 3],[4 1]));
% w = 1/4, 3/2 (so the range is from 1e-1 to 1e1), k = 5;
w = [1e-2, 1/4, 3/2, 1e2]; % vector that contains the frequencies in the ascending
order + range margins
s = [20, 0, -20]; % vector with the slopes after each corner frequency
k = 5;
wl = logspace(-2, 2, 1e2); % margins for the bode plot
bode_approx(H, k, w, s, wl);
```

```
Current plot held
Current plot released
```


Bode Diagram
Magnitude (dB)
5
0
-5
-10
-15
-20
-25
-30
-35
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Frequency (rad/s)